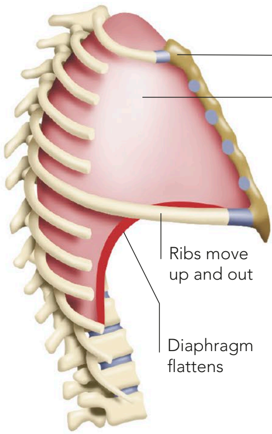
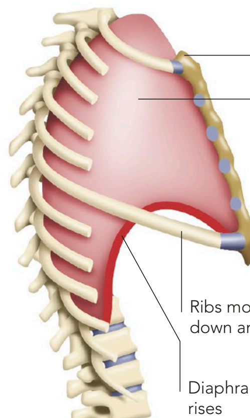
VOLUME AND PRESSURE

Breathing alters the volume of the chest (thoracic cavity). The lungs "suck" on to the inner chest wall, so that as the cavity expands, they also become larger. The main expanding forces are provided by the diaphragm and intercostal muscles. At rest, the diaphragm carries out most of the work, as 0.5 litres (17fl oz) of air – the tidal volume – shifts in and out with each breath (12 to 17 times every minute). Rate and volume increase automatically if the body needs more oxygen, as during exercise. Then forced inspiration can suck in an extra 2 litres (70fl oz), and forced expiration expels almost as much, leading to a total air shift, or vital capacity, of more than 4.5 litres (150fl oz) in a large, healthy adult. The breathing rate can triple, producing a total air exchange more than 20 times greater than at rest.



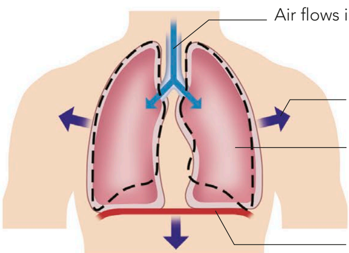
BREATHING IN

The diaphragm contracts to become less dome-like, while the ribs swing upward and outward with a "bucket handle" action to raise the sternum.



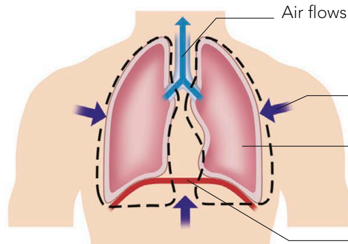
BREATHING OUT

The diaphragm relaxes, and the elastic, stretched lungs recoil to become smaller again, allowing the sternum and ribs to move down and inward.



NEGATIVE PRESSURE

As the lung volume increases, the air pressure within decreases. Atmospheric pressure outside the body is now higher, and air is drawn down the airways and into the lungs – in effect, air is "sucked" in.



POSITIVE PRESSURE

As the lung volume diminishes when exhaling, the air is compressed, which raises its pressure within the lungs. So the air is pushed back along the airways and out of the nose and mouth.